

GABES, Tunisia

WMO: 607650

Lat: 33.877N

Lon: 10.103E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.3	41.4	20.9	15.7	63.2	24.4	18.5	57.5	29.1	58.8	26.1	58.1	5.7	270	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.0	103.6	69.6	98.6	70.0	94.8	70.9	80.9	88.1	79.6	87.0	78.2	86.1	11.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	149.8	86.4	77.2	141.9	85.2	75.6	134.3	84.2	44.5	88.2	43.1	87.0	41.6	86.2	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.9	21.7	19.1	DB	35.8	111.8	2.4	2.9	34.0	113.9	32.6	115.6	31.3	117.2	29.6	119.3
			WB	32.4	82.7	2.2	2.7	30.9	84.6	29.6	86.2	28.4	87.8	26.8	89.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	53.6	55.4	60.4	65.9	72.0	78.3	83.3	84.3	80.3	73.2	63.2	55.7	(6)
	DBStd	11.75	4.03	4.99	5.23	4.71	4.94	4.81	4.05	3.59	4.62	5.10	4.94	4.51	(7)
	HDD50	23	11	7	0	0	0	0	0	0	0	0	0	5	(8)
	HDD65	1216	354	273	164	41	2	0	0	0	0	2	90	290	(9)
	CDD50	6917	122	158	322	477	682	850	1032	1063	909	721	398	183	(10)
	CDD65	2630	0	4	20	67	218	400	567	598	459	258	37	2	(11)
	CDH74	27877	2	47	256	531	1492	3831	7044	7664	4833	1939	221	17	(12)
	CDH80	12350	0	10	89	185	595	1585	3431	3835	1966	615	37	2	(13)
(14) Wind	WSAvg	9.2	9.1	9.1	9.4	10.0	10.7	10.5	9.8	9.2	9.2	7.7	7.9	8.4	(14)
(15) Precipitation	PrecAvg	7.9	0.8	0.6	0.8	0.6	0.3	0.1	0.0	0.0	0.9	1.4	0.9	1.1	(15)
	PrecMax	16.9	4.8	3.6	3.6	4.8	1.1	0.6	0.3	0.7	4.8	10.5	4.2	7.9	(16)
	PrecMin	3.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	3.2	1.0	0.8	0.8	0.8	0.3	0.2	0.1	0.1	1.1	2.0	1.1	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	80.4	89.6	93.2	102.1	107.9	109.6	108.6	104.1	96.8	84.4	76.6	(19)
		MCWB	54.0	56.1	60.6	60.9	65.0	67.7	69.8	70.5	69.6	65.2	60.8	56.0	(20)
	2%	DB	67.7	73.3	80.7	85.6	92.0	98.6	102.4	100.8	97.2	91.0	79.2	69.9	(21)
		MCWB	52.9	54.5	56.0	60.7	64.5	67.5	71.9	71.8	71.0	66.6	61.0	55.4	(22)
	5%	DB	64.6	68.4	75.0	79.6	86.0	91.8	96.8	96.6	92.7	86.2	76.1	66.8	(23)
		MCWB	51.9	52.5	55.7	60.2	64.7	68.2	71.2	72.5	71.7	66.9	61.0	54.9	(24)
	10%	DB	62.3	64.8	70.2	75.3	81.0	87.8	92.9	92.9	88.5	82.5	73.2	64.5	(25)
		MCWB	51.2	51.7	55.4	59.6	63.9	68.5	71.9	73.2	73.2	68.6	60.4	54.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	59.3	63.9	69.1	73.0	78.2	82.2	82.9	81.8	77.3	70.0	62.3	(27)
		MCDB	63.0	67.4	74.1	76.8	82.3	86.1	88.7	90.8	88.5	82.1	75.4	66.1	(28)
	2%	WB	56.6	57.5	61.6	66.4	71.1	76.2	80.0	81.2	79.7	75.1	67.1	59.9	(29)
		MCDB	60.6	65.3	69.0	74.0	79.3	84.9	87.5	88.9	86.2	81.2	72.7	64.6	(30)
	5%	WB	55.2	55.8	60.2	64.5	69.6	74.6	78.5	80.0	78.1	73.3	64.8	57.9	(31)
		MCDB	60.4	63.1	67.4	71.8	77.5	83.1	86.5	87.7	85.2	79.5	71.4	63.7	(32)
	10%	WB	53.6	54.3	58.8	63.0	68.2	73.0	77.1	78.8	76.6	71.5	62.5	56.0	(33)
		MCDB	59.9	61.9	65.8	70.4	75.9	81.6	85.8	86.6	84.1	78.1	69.4	62.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.0	18.0	17.1	16.3	16.8	17.5	17.0	16.2	17.2	17.8	17.2	(35)
		MCDBR	20.8	23.6	27.5	26.6	27.1	26.6	27.6	26.2	24.4	23.9	22.1	19.9	(36)
	5% WB	MCWBR	11.7	11.5	11.9	11.3	10.7	10.7	11.9	11.9	11.6	10.7	11.5	11.5	(37)
		MCWBR	16.5	17.7	17.4	17.3	15.8	16.2	16.3	15.4	15.6	15.2	16.2	16.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.369	0.398	0.484	0.536	0.553	0.510	0.537	0.525	0.530	0.482	0.425	0.380	(40)
		taud	2.241	2.144	1.912	1.813	1.801	1.928	1.882	1.917	1.910	1.992	2.093	2.202	(41)
	Ebn at Noon		260	265	250	243	240	250	243	243	234	233	236	247	(42)
		Edh at Noon	35	43	58	67	69	61	63	60	58	49	40	34	(43)
(44) All-Sky Solar Radiation	RadAvg		981	1267	1619	1965	2216	2413	2421	2160	1738	1355	1040	882	(44)
	RadStd		46	53	79	87	110	77	63	61	67	62	41	56	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-2.43	-3.64	+4.93	-1.51	-1.78	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	+0.40	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	+185	+108	(47)

Nomenclature: See separate page